

1)

Ethanol (C_2)

~ LNK

i-propanol (iC_3 , b.p = $83^\circ C$)

~ LK

n-propanol (nC_3 , b.p = $98^\circ C$)

~ HK

n-butanol (C_4)

i-propanol (iC_3) ~ LK

Excellent!

①

5/5

2)

5/5

$F = 170 \text{ kmol/h}$

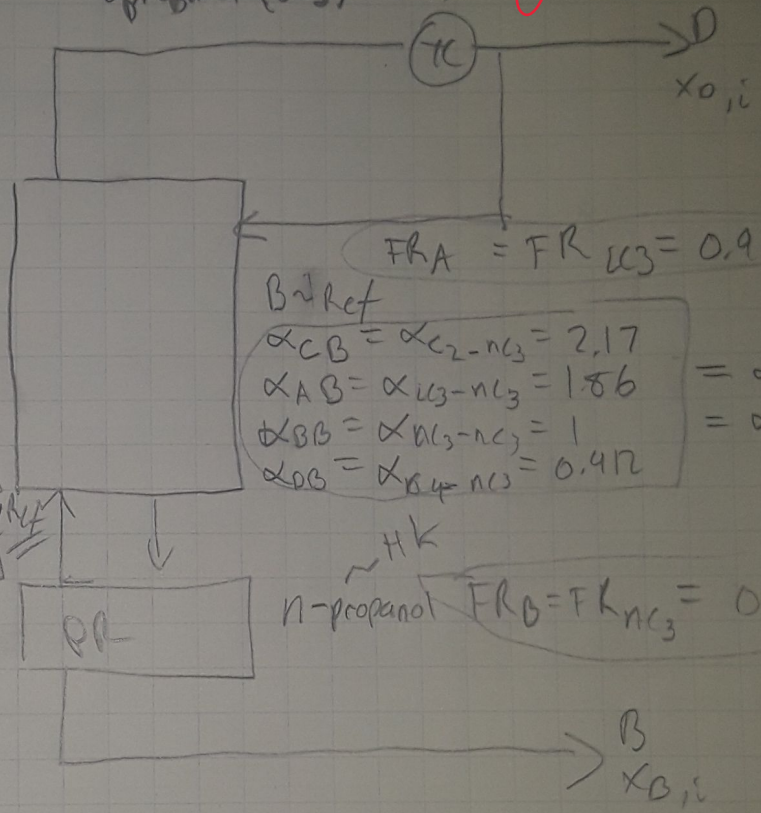
$q = 0$
Sat. vapor

C_2 Ethanol: $0.25 = z_{C_2}$ [C]

iC_3 i-propanol: $0.25 = z_{iC_3}$ [A]

nC_3 n-propanol: $0.30 = z_{nC_3}$ [B]

C_4 n-butanol: $0.20 = z_{C_4}$ [D]



$FRA = FR_{iC_3} = 0.98$

But Ref

$\alpha_{CB} = \alpha_{C_2-nC_3} = 2.17$

$\alpha_{AB} = \alpha_{iC_3-nC_3} = 1.86$

$\alpha_{BB} = \alpha_{nC_3-nC_3} = 1$

$\alpha_{DB} = \alpha_{C_4-nC_3} = 0.412$

= α_{LK-HK}
= α_{HK-HK}

~ HK

n-propanol $FR_B = FR_{nC_3} = 0.987$

3) MB: $In = Out$

$\rightarrow F = D + B$

5/5

$(MB) \rightarrow Fz_i = Dx_{D,i} + Bx_{B,i}$

Note: species i changes for the 4 components

$FR_{i,D} = \frac{Dx_{D,i}}{Fz_i}$

4) Assume all LNK go to bottom LNK $\rightarrow$ Distillate
all LNK go to top

Distillate LNK $\rightarrow Dx_{C_2} = Fz_{C_2}$

$= (170)(0.25) = 42.5$

LK $\rightarrow Dx_{iC_3} = (FR_{iC_3})(Fz_{iC_3}) = (0.98)(170 \cdot 0.25)$

$\Rightarrow \begin{bmatrix} Dx_{C_2} = 42.5 \\ Dx_{iC_3} = 41.65 \end{bmatrix} \text{ kmol/h}$

Dx_i ☒

4 cont)

②

HW : $DX_{nc3} = (1 - FR_{nc3, Bot}) (Fz_{nc3}) = (1 - 0.987)(170)(0.30)$
 HWK : $DX_{cy} = 0$ Assumption

$\Rightarrow \left[\begin{matrix} DX_{nc3} = 0.663 \\ DX_{cy} = 0 \end{matrix} \right] \text{ kmol/h } DX_i \checkmark$

$D = \sum DX_E = 42.5 + 41.65 + 0.663 + 0$
 $\left[D = 84.813 \text{ kmol/h} \right] \checkmark D \checkmark$

$\left[X_E = \frac{\sum DX_E}{D} \right]$

20/20

$X_{c2, dis} = \frac{42.5}{84.813} = 0.5011$

$X_{c3, dis} = \frac{41.65}{84.813} = 0.491$

$X_{nc3, dis} = \frac{0.663}{84.813} = 0.007817$

$X_{c4, dis} = 0$

check $\sum X_E = 0.99999$
 $\sum X_E \approx 1$

5) Bottoms HWK, HWK, LW
 LNK : $BX_{c2, Bot} = 0$ Assumption

LK : $BX_{c3, Bot} = (1 - FR_{c3, Bot}) (Fz_{c3}) = (1 - 0.987)(170 \cdot 0.25)$

HW : $BX_{nc3, Bot} = (FR_{nc3}) (Fz_{nc3}) = (0.987)(170 \cdot 0.30)$

HWK : $BX_{cy, Bot} = Fz_{cy} = (170 \cdot 0.20)$

$Fz = BX_E$ or $FR_B = \frac{BX_E}{Fz}$

$\left[\begin{matrix} BX_{c2, Bot} = 0 \\ BX_{c3, Bot} = 0.85 \\ BX_{nc3, Bot} = 50.337 \\ BX_{cy, Bot} = 34 \end{matrix} \right] \text{ kmol/h } BX_i \checkmark$

check $F = D + B$
 $B + D = 170$

$B = \sum BX_E = 0 + 0.85 + 50.337 + 34$
 $\left[B = 85.187 \right] \text{ kmol/h } B \checkmark$

$$X_{C, Bot} = \frac{\sum B X_{E, Bot}}{B}$$

$$X_{C2, Bot} = 0 = 0$$

$$X_{C3, Bot} = \frac{6.85}{85.187} = 0.009928$$

$$X_{C3, Bot} = \frac{60.337}{85.187} = 0.5909$$

$$X_{C4, Bot} = \frac{34.5}{85.187} = 0.399$$

$$(check) \sum X_C \approx 1 = 1$$

20/20

$X_{C, B}$ ✓

$$6) N_{min} = \frac{\ln \left[\frac{(F_{KA, dist})(F_{KB, Bot})}{(1-F_{KA})(1-F_{KB})} \right]}{\ln \alpha_{AB}}$$

$$= \frac{\ln \left[\frac{(0.98)(0.987)}{(1-0.98)(1-0.987)} \right]}{\ln (1.86)}$$

$$F_{KA, dist} = 0.98$$

$$F_{KB, Bot} = 0.987$$

$$\alpha_{AB} = X_{C3, Bot} = 1.86$$

$$N_{min} = 13.248$$

$$u_3 = A - LK \quad n_3 = B - HK$$

$$N_{F, min} = \ln \left[\frac{\left(\frac{X_{LK}}{X_{HK}} \right)_{dist}}{\left(\frac{Z_{LK}}{Z_{HK}} \right)} \right]$$

$$Bot: X_{LK} = X_{C3, Bot} = 0.491$$

$$X_{HK} = X_{C3, dist} = 0.007017$$

$$Z_{LK} = 0.25$$

$$Z_{HK} = 0.30$$

$$= \ln \left[\frac{\left(\frac{0.491}{0.007017} \right)}{\left(\frac{0.25}{0.30} \right)} \right]$$

$$\ln (1.86)$$

$$\alpha_{LK-HK} = \alpha_{AB} = 1.86$$

$$N_{F, min} = 6.965$$

$$N_{F, min}$$

$$check = \ln \left[\frac{P_{LK}}{O_{LK}} \right] \frac{Z_{LK}}{Z_{HK}}$$

$$\ln \alpha_{AB}$$

7) $\Delta V_{\text{feed}} = F(1-q)$ sat. vapor $\Rightarrow q=0$
 $= F(1-0)$ ✓

$\Delta V_{\text{feed}} = F = 170 \text{ kmol/h}$ ✓

ΔV_{feed} ✓

5/5

8) $V_{\text{min}} = \sum_{i=1}^C \frac{x_{i-\text{ref}}(Dx_{i,\text{dist}})}{x_{i-\text{ref}} - \phi}$

$\phi = 1.67$

$(D_{\text{cu,dis}} = 0)$

$V_{\text{min}} = \frac{(x_{\text{C2-nc3}})(Dx_{\text{C2}})}{x_{\text{C2-nc3}} - \phi} + \frac{(x_{\text{C3-nc3}})(Dx_{\text{C3}})}{x_{\text{C3-nc3}} - \phi} + \frac{(x_{\text{nc3-nc3}})(Dx_{\text{nc3}})}{x_{\text{nc3-nc3}} - \phi} + 0$

$= \frac{(2.17)(42.51)}{2.17 - 1.67} + \frac{(1.86)(41.65)}{1.86 - 1.67} + \frac{1.0(0.663)}{1 - 1.67}$

$= 184.65 + 407.73 + (-6.989)$

$V_{\text{min}} = 591.192 \text{ kmol/h}$ ✓

V_{min} ✓

MB: $I_{\text{in}} = \text{out}$

$V_{\text{min}} = D + L_{\text{min}}$

$L_{\text{min}} = V_{\text{min}} - D = 591.192 - 84.813$

$L_{\text{min}} = 506.379 \text{ kmol/h}$

$\left(\frac{L}{D}\right)_{\text{min}} = \frac{506.379}{84.813} = 5.9705$

$\left[\left(\frac{L}{D}\right)_{\text{min}} = 5.9705\right]$

$\left(\frac{L}{D}\right)_{\text{min}}$ ✓

15/15

$$a) \frac{L}{D} = 1.10 \left(\frac{L}{D} \right)_{\min} = 1.10 (5.9705)$$

⑤

$$\left[\frac{L}{D} = 6.5675 \right]$$

$$\Rightarrow X = \frac{(L/D) - (L/D)_{\min}}{(L/D) + 1} = \frac{(6.5675) - (5.9705)}{6.5675 + 1}$$

$$\left[X = 0.07888 \right]$$

0.01 < X < 0.9
→ 2nd corr.

$$\Rightarrow y = 0.545827 - 6.591422X + \frac{6.002743}{X}$$

$$\left[y = 0.5339 \right]$$

$$\Rightarrow y = \frac{N - N_{\min}}{N + 1}$$

$$yN + y = N - N_{\min}$$

$$y + N_{\min} = N - yN$$

$$y + N_{\min} = N(1 - y)$$

$$\Rightarrow N = \left[\frac{y + N_{\min}}{1 - y} \right] = \frac{(0.5339) + (13.248)}{1 - 0.5339}$$

$$\left[N = 29.5656 \right]$$

N ✓

15/15

$$\Rightarrow \frac{N_{F, \min}}{N_{\min}} = \frac{N_F}{N}$$

$$N_F = \frac{N_{F, \min}}{N_{\min}} N = \left(\frac{6.965}{13.248} \right) (29.5656)$$

$$\left[N_F = 15.5454 \right]$$

N_F ✓